

DP Computer Science: A 1.1.2 GPU Architecture

1. Core Concepts & Learning Objectives

Learning Intentions

By the end of this lesson, students will be able to:

1. Describe the role of a **GPU** as a co-processor that applies one operation to many data items at the same time
 2. Name the architectural features that make a GPU suitable for parallel work: **many simple cores**, **parallel execution**, and **high-bandwidth dedicated memory**
 3. Describe how **graphics rendering** and at least one **non-graphics use** rely on those features
 4. For a given real-world scenario, identify the **parallel workload** and describe the **architectural feature** that handles it
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Key Vocabulary (Tier 3)

Term	Definition
GPU (Graphics Processing Unit)	A co-processor that applies the same operation to very many data items at once
Co-processor	A specialised processor that takes particular work off the main CPU
Parallel execution	Performing many calculations in the same moment rather than one after another
Core	A processing unit inside a chip; a GPU has hundreds or thousands of simple ones
Memory bandwidth	How much data can be moved to and from memory each second

Term	Definition
GPGPU	Using a GPU for non-graphics work that suits parallelism

Pedagogical Focus

- **Communication (ATL):** Students consciously practise describing precisely—connecting a named workload to a named architectural feature in full sentences using accurate vocabulary.
- **Thinking Skills (ATL):** Students identify what different real-world scenarios have in common.
- **Social Skills (ATL):** Students work in pairs to articulate their understanding.
- **Global Contexts (ATT):** The lesson anchors concepts in diverse settings—video games, weather simulation, AI, and cryptocurrency mining—with ethical discussion around energy costs.

EAL Scaffold

Support Type	Description
Sentence Frame	"A GPU is suited to [scenario] because the work involves [the parallel workload, e.g., millions of pixels, matrix operations, or grid cells], which its [architectural feature] processes in parallel."
Idiom Flag	Replace "Explain it like I'm 5" with "Explain it in simple words, as if to someone who has never studied computing."

2. Lesson Sequence (One-Hour Block)

Part 1: Do Now (Retrieval) — ~10%

Activity: Four questions recalling A1.1.1 (CPU components), displayed on the board as students enter:

1. What are the two main CPU units and their jobs?
2. What is the role of the program counter?

3. What is a co-processor?
4. What is one difference between single-core and multi-core processors?

Land the co-processor item since the GPU is the co-processor students meet today.

Part 2: Main Sequence — ~60%

Phase A: Anchor — Games (Slides 5–8)

- Ask: *"Which of Pac-Man, Minecraft, and Cyberpunk 2077 needs more GPU power, and why?"*
 - Move briskly through full-screen images
 - Draw out: **more visual detail = more pixels and effects to compute**
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Phase B: Anchor — Four Domains (Slides 9–12)

Show four scenarios and ask: *"What could these possibly share?"*

Scenario	Workload Type
 Video Game	Millions of pixels per frame
 Weather Simulation	Millions of independent grid cells
 AI Training	Vast numbers of matrix operations
 Cryptocurrency Mining	Billions of hashing calculations

Keep the question open—students can judge games by eye but not the others, so the focus shifts from scale to shared principle.

Phase C: Teach Through the Scenario (Slides 13–19)

Direct Instruction: The GPU as a co-processor:

- Working principle: **one instruction over many data items**
- Three architectural features:
 1. **Many simple cores** (hundreds or thousands)

2. **Parallel execution** (calculations happen simultaneously)
3. **High-bandwidth dedicated memory** (data moves quickly)

Model the two-part move throughout: *Name the workload → then the feature that handles it.*

Scenario	Parallel Workload	Architectural Feature
Video Games	Millions of pixels per frame	Many cores in parallel; high-bandwidth memory
Weather Simulation	Same calculation over millions of grid cells	Cores computing at once
AI Training	Matrix operations split into independent calculations	Parallel execution across cores
Cryptocurrency Mining	Same hashing calculation done billions of times	Parallel execution; non-graphics use case

Phase D: Return (Slide 20)

- Go back to the four uses
- Students describe each in the **workload-then-feature** form
- Close with the **energy and ethics prompt** as a short discussion (not a debate)
 - *Energy cost of cryptocurrency mining and large-model training*

Part 3: Deliberate Practice — ~20%

Activity (Pairs):

- Each pair takes **one scenario**:
 - Video game frame
 - Weather simulation
 - Training a neural network
 - Cryptocurrency mining
- Write a short description using the **sentence frame**
- Say it aloud to another pair who check that **both the workload and feature were named**

Teacher Role: Spread scenarios across the room. Listen for the link, not extra detail.

| **Weak answer:** "A GPU is fast or good at graphics." → **Prompt:** "Which feature, and over what data?"

Part 4: Exit Check — ~10%

Written Prompt:

| "Describe the role of a GPU, naming one architectural feature and one real-world scenario it suits."

Assessment: Mark for the link, not length.

Additional Check: Use questions 1, 3, and 6 from the eBook quiz as hinge questions:

- A wrong answer on 3 or 6 signals the workload-to-feature link has not landed → reopen at the start of the next lesson

Close: Return to the communication skill named at the start.

3. Assessment & Exemplar Responses

Exemplar Descriptions

Scenario	Exemplar Response
Video Games	"A GPU is suited to video games because the work involves millions of pixels per frame, which its many cores process in parallel with high-bandwidth memory supplying them."
Weather Simulation	"A GPU is suited to weather simulation because the work involves the same calculation over millions of independent grid cells, computed across the cores at once."
AI Training	"A GPU is suited to AI training because the work involves vast numbers of matrix operations that split into independent calculations, run in parallel."

Scenario	Exemplar Response
Cryptocurrency Mining	"A GPU is suited to cryptocurrency mining because the work involves the same hashing calculation done billions of times, run in parallel; this is not graphics, which is the point."

4. Differentiation & Extension

Support (Scaffolding That Can Be Removed)

Support Strategy	Description
Sentence Frame Card	Provide for paired task; withdraw for exit check
Half-Completed Worked Example	Workload named, feature left blank for pairs who need a way in
Annotated Board Photo (Slide 16)	Concrete referent for students who find architecture hard to visualise

Challenge

Challenge Strategy	Description
Wrong Choice Analysis	Ask stronger pairs to take a case where a GPU is the wrong choice (e.g., a single complex sequential calculation) and describe why the same architecture that helps with parallel work does not help here
Preparation for HL	Stays at description but prepares the ground for the HL comparison in A1.1.3

5. Engagement Activities (Supplementary)

If time allows, these can replace or supplement the main sequence:

Activity	Description
"Guess the Game"	Show screenshots/clips of games with varying graphical complexity; discuss which needs more GPU power and why
Interactive Simulation	Use GPUSim, Shadertoy, or WebGL tools to experiment with GPU configurations and observe performance impacts
Collaborative Research	Groups research a GPGPU application and present findings
"Explain in Simple Words"	Groups explain a GPGPU application as if to a non-specialist
GPU vs. CPU Debate	Teams debate pros and cons of each for specific tasks
"Design a GPU" Challenge	Hypothetical scenario: design a GPU for a new gaming console or self-driving car

6. Resources

- **Presentation** (Slide Deck)
- **InThinking eBook** (A 1.1.2)
- **GPU Simulators:** GPUSim, Shadertoy, WebGL
- **Quiz:** 10 multiple-choice questions from eBook

7. Archived Lesson Plan (Previous Format)

This section preserves the earlier structure for reference.

Learning Objectives (Archived)

- Explain the role of GPUs in computer systems
- Describe the key architectural features of GPUs
- Identify applications of GPUs beyond graphics rendering
- Evaluate the advantages of using GPUs for specific tasks

Archived Activity Sequence

Phase	Time	Activity
Engage	10 mins	"Guess the Game" — screenshots of Minecraft vs. Cyberpunk 2077
Explore	20 mins	Interactive simulation + collaborative research on GPGPU applications
Explain	15 mins	"Explain in simple words" + Visual presentation with diagrams
Elaborate	20 mins	"GPU vs. CPU" debate + "Design a GPU" challenge
Evaluate	15 mins	Hinge-question quiz (10 MCQs) + Exit ticket paragraph

Author's Reflection

"I have kept the parts of the original presentation that work, such as the architecture slides and the spread of applications, because they describe the GPU accurately and at the right level. The previous version leaned on external tools in the explore phase, but those did not give immediate access to the configurations I wanted students to see, and in a single hour that cost more time than it returned. I have replaced open exploration with a short teacher-led demonstration and reserved genuine investigation for later lessons in the unit. The old explain, elaborate and evaluate sequence was loose, so I have rebuilt the lesson around real scenarios: students first meet four different uses—a video game, a weather simulation, AI, and cryptocurrency mining—then learn the one architectural idea that connects them, then return to describe each in those terms.

Starting with uses that look unrelated is deliberate, because the point students should leave with is that the same parallel design serves all of them. I have held the lesson at the command term 'Describe'—meaning here linking a workload to an architectural feature, not weighing the GPU against the CPU, which is the HL statement A1.1.3 and now carries the comparison work that used to sit on this page."